

Antonio Nikoloski

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SUMMARY

Computer Science student focused on building efficient, real-world software systems. Hands-on experience in GPU computing, data processing, and AI integration. Quick to pick up new technologies, comfortable in team environments, and committed to solutions that perform under pressure.

PROJECTS

CUDA-Accelerated Genome Alignment System

Python · CUDA · CuPy · NumPy · Matplotlib

- GPU-accelerated implementation of the Smith–Waterman DNA sequence alignment algorithm
- Designed custom CUDA kernels with optimized memory access for maximum parallel throughput
- Achieved 2,000×+ speedup over CPU baselines on large genomic datasets
- Validated correctness across diverse mutation scenarios and edge cases

Multi-Agent Hide-and-Seek Simulation

Python · Pygame · A Pathfinding · AI Behavior Modeling*

- Built a tile-based simulation with two intelligent agents (Hider & Seeker) using A* navigation
- Implemented energy-based behavior systems and dynamic state transitions for realistic agent logic
- Designed procedural environments and agent perception systems for varied, emergent interactions

AI-Powered Learning Assistant Platform

Django · Python · LLM Integration | In Progress

- Building a backend that ingests lecture content and generates simplified explanations via LLMs
- Designing automatic flashcard and quiz generation pipelines for student study workflows
- Architecting for scalable AI integration from the ground up

Educational Web Platform for Children with Special

Needs *HTML · SCSS · JavaScript · Responsive Design*

- Designed and developed a fully responsive educational website optimized for accessibility•
- Prioritized intuitive UX for users with varying cognitive and motor needs

TECHNICAL SKILLS

Languages	JavaScript, Python, Java, C/C++, SQL, HTML/CSS
Frameworks & Tools	React, Node.js, Django, Git, Docker, AWS, GitHub Actions
Libraries	CUDA, CuPy, NumPy, Matplotlib
Core Concepts	Parallel Computing, Machine Learning, Algorithms, Data Analysis
Languages (Spoken)	Macedonian (Native), English, Serbian

EDUCATION

Ss. Cyril and Methodius University — Skopje

B.Sc. Computer Science · GPA: 7.9 / 10

Sep 2022 – Present

High School "Nace Bugjoni" — Kumanovo

Mechanical Engineering Technician · GPA: 5.0 / 5.0 · Top Student, Mechanical Engineering

2018 – 2022